Recap: Expectation and Variance

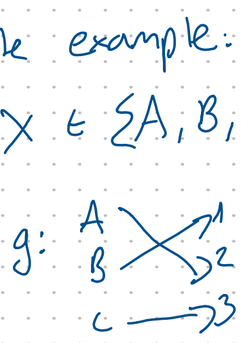
Expectation

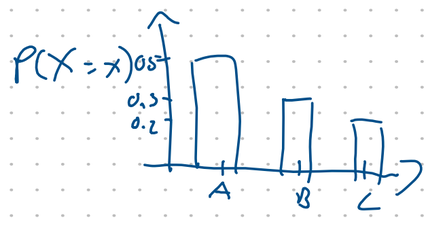
- Continuous RV X:
  $\mathbb{E}[g(X)] = \int_{-\infty}^{\infty} g(x) \cdot p(x)\, dx$
  where $p(x)$ is the pdf of X
- Discrete RV X:
  $\mathbb{E}[g(X)] = \sum_x g(x) \cdot P(X=x)$

Intuition: weight $g(x)$ with the probabilities of X

Simple example:

$X \in \{A, B, C\}$





$$\mathbb{E}[g(X)] = \sum_{x=A}^{C} g(x) \cdot P(X=x)$$
$$= g(A) \cdot P(X=A) + \dots$$
$$= 2 \cdot 0.5 + 1 \cdot 0.3 + 3 \cdot 0.2 = 1.9$$

Variance

$$\mathrm{Var}(g(X)) = \mathbb{E}\left[\left(g(X) - \mathbb{E}[g(X)]\right)^2\right]$$

probability-weighted squared deviation from expected value

Joint & Conditional

$\mathbb{E}_X$ / $\mathbb{E}_Y$

$\mathbb{E}_{Y|X=x}$

$\mathbb{E}_{XY}$

} subscript says over which distribution we average

$\mathbb{E}_{X \sim \mathbb{P}_X}$ (marginal)

$\mathbb{E}_{Y \sim \mathbb{P}_{Y|X=x}}$ (conditional at $X=x$)

$\mathbb{E}_{X,Y \sim \mathbb{P}_{XY}}$ (joint)

Example:

Joint

X \ Y	1	3	4	6
0	1/4	1/4	0	0
1	0	0	1/4	1/4

$P(Y=y)$ $= \frac{1}{4}+0$ $= 1/4$ | 1/4 | 1/4 | 1/4 $\rightarrow$ $\mathbb{E}_Y[Y] = 1 \cdot 1/4 + 3 \cdot 1/4 + 4 \cdot 1/4 + 6 \cdot 1/4$
$= 3.5$

marginal view $\rightarrow$ "looking from the side" $\rightarrow$ each Y is equally likely

$P(Y|X=0)$: conditional view $\rightarrow$ "looking at a slice" / "we know that $X=0$"

$P(Y=y|X=0) = P(Y=y, X=0) / P(X=0)$

$\rightarrow$ divide each value by $P(X=0) = 1/2$ to renormalize

$1/2$ $1/2$ 0 0 $\rightarrow$ $\mathbb{E}_{Y|X=0}[Y] = 1 \cdot 1/2 + 3 \cdot 1/2 + 4 \cdot 0 + 6 \cdot 0$
$= 2$

$\mathbb{E}_{Y|X=1}[Y] = 5$